

ROS node topic information

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Node Name

Robot publishes topics

Robot subscription topic

Car test control

Control car movement

Control buzzer

Control PWM servo

Node Name

/YB_Car_Node

Robot publishes topics

Function	Topic	Information	Frame_id	Frequency
Car speed	/odom_raw	nav_msgs/msg/Odometry	odom_frame	11 Hz
IMU chip	/imu	sensor_msgs/msg/Imu	imu_frame	25 Hz
LiDAR	/scan	sensor_msgs/msg/LaserScan	laser_frame	12 Hz
Battery voltage	/battery	std_msgs/msg/UInt16		1Hz

Notes:

1. The actual battery voltage is the data value of the /battery topic divided by 10.0, in V. For example, if data=82, the actual voltage is 8.2V.

Robot subscription topic

Function	Topic	Information
Control car	/cmd_vel	geometry_msgs/msg/Twist
Control buzzer	/beep	std_msgs/msg/UInt16
Control servo S1	/servo_s1	std_msgs/msg/Int32
Control servo S2	/servo_s2	std_msgs/msg/Int32

Notes:

1. The data value of the control buzzer is [0, 10000]. When data=0, the buzzer is turned off. When data=1, the buzzer keeps beeping. When data>=10, the buzzer automatically turns off

after data milliseconds.

2. The data value of the control servo S1 is [-90, 90].

3. The data value of the control servo S2 is [-90, 20].

Car test control

Control car movement

Publish data to the /cmd_vel topic to control the robot car to move forward at 0.5m/s.

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.5, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Publish data to the /cmd_vel topic to control the robot car to rotate at 1.5 rad/s.

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.5}}"
```

Publish data to the /cmd_vel topic to control the robot car to stop.

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Control buzzer

Publish data to the /beep topic to control the buzzer to keep beeping.

```
ros2 topic pub --once /beep std_msgs/msg/UInt16 "data: 1"
```

Publish data to the /beep topic to turn off the buzzer.

```
ros2 topic pub --once /beep std_msgs/msg/UInt16 "data: 0"
```

Publish data to the /beep topic to control the buzzer to sound for 300 milliseconds and then turn off automatically.

```
ros2 topic pub --once /beep std_msgs/msg/UInt16 "data: 300"
```

Control PWM servo

Publish data to the /servo_s1 topic to control servo S1 to rotate to 30°.

```
ros2 topic pub --once /servo_s1 std_msgs/msg/Int32 "data: 30"
```

Publish data to the /servo_s2 topic to control servo S2 to rotate to -60°.

```
ros2 topic pub --once /servo_s2 std_msgs/msg/Int32 "data: -60"
```

